

PROJET D'ÉTUDES ET PROJET PROFESSIONNEL

(STATEMENT OF PURPOSE)

Master of Science in Computer Science: Software Architecture, Distributed Systems & Cloud Infrastructure

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Target Intake: Fall 2027 (Academic Year 2027/2028)

Portal Dossier: Campus France Éthiopie / Études en France

Fellowship Candidacy: France Excellence Eiffel Scholarship (*Bourse d'Excellence Eiffel*)

1. Introduction & Academic Ambition

Software systems today are the invisible infrastructure underpinning global commerce, public governance, and economic resilience. However, building software that remains mathematically correct, highly available, and resilient under extreme concurrency is one of the most demanding frontiers in computer science. As an Ethiopian software engineer with a Bachelor of Science in Computer Science (Major GPA: 3.20/4.00) from St. Mary's University and proven production experience designing geospatial engines and fault-tolerant financial state machines, my objective is to pursue an advanced **Master of Science (M1/M2) in Computer Science with a specialization in Distributed Systems, Software Architecture, and Cloud Infrastructure in France**.

France stands at the vanguard of European algorithmic and computational science—pioneering foundational research at Inria, LIP6, and CNRS while fostering a world-renowned software engineering ecosystem. Through this Master's degree, I intend to bridge my rigorous hands-on engineering background with elite European theoretical frameworks, preparing myself to design dependable, high-integrity digital infrastructure capable of scaling across complex, resource-constrained environments.

2. Academic Foundations & Computational Rigor

My intellectual dedication to computer science was cultivated during my undergraduate studies at St. Mary's University in Addis Ababa (conducted entirely in English). Across four rigorous years, I maintained a **Major GPA of 3.20 / 4.00**, demonstrating sustained academic excellence across core computational disciplines:

- **Algorithms & Data Structures:** Developing algorithmic thinking, asymptotic complexity analysis, graph algorithms, and dynamic programming.
- **Operating Systems & Systems Programming:** Analyzing process scheduling, thread synchronization primitives, virtual memory hierarchies, and low-level inter-process communication (IPC).

- **Database Management Systems & Relational Theory:** Mastering 3NF normalization, ACID transaction guarantees, index structures (B+ trees), query execution planners, and concurrency conflict resolution.
- **Distributed Networks & Protocols:** Gaining end-to-end understanding of TCP/IP socket architectures, network topology, and transport security.

To augment my undergraduate curriculum with modern quantitative finance, I completed an intensive 50-hour **Executive Certification in Capital Markets and Securities Analysis** at Addis Ababa University's School of Commerce in collaboration with the Ethiopian Capital Market Authority (September 2024). This training deepened my understanding of market microstructure, transactional clearance mechanisms, and systemic financial risk—principles that now guide how I architect fault-tolerant distributed ledger systems.

3. Engineering Rigor & Production Architecture

Unlike applicants whose knowledge remains purely theoretical, I have engineered and deployed complex software solutions that solve tangible operational and infrastructural challenges:

1. Addis Condo Finder (Geospatial Distributed Property Engine):

- *The Problem:* Urban property search in Addis Ababa suffered from non-standardized geolocation and unverified spatial data.
- *The Solution:* I architected a full-stack platform using Flutter (mobile frontend) and a Laravel 13 REST API backend backed by spatial database indexing. To ensure rapid mobile rendering over intermittent 3G/4G connectivity, I designed bounding-box viewport queries (`bbox=minLat,minLng,maxLat,maxLng`) that restrict payload delivery strictly to the visible map viewport.
- *Resilience & Consensus:* I implemented client-side token-matching search across condominium names and block numbers with sub-50ms query response times, coupled with write-rate limiters and atomic unique constraints to eliminate multi-user verification race conditions.

2. LuckyDraw High-Concurrency Financial Gaming & Ledger Engine:

- *The Problem:* High-traffic gaming and lottery engines frequently suffer from double-spend vulnerabilities, double-payout anomalies, and floating-point ledger inaccuracies under concurrent user load.
- *The Solution:* I engineered an atomic backend engine in Laravel 13 and MySQL InnoDB using strict pessimistic row-level locking (`lockForUpdate`). This mathematically guarantees that jackpot draws, ticket reservations, and prize allocations execute in strict serial isolation, entirely preventing double-payouts.
- *Accounting Integrity:* Balances and transactions are computed in integer cents (`tests/Unit/MoneyTest.php`), eliminating floating-point rounding errors. I validated this architecture with an automated test suite of over 25 unit, feature, and concurrency stress tests.

3. Money Manager (Offline-First Reactive Personal Finance System):

- *The Problem:* Users in developing nations require privacy-preserving, offline-first personal finance tracking without cloud latency or dependency on active connectivity.
 - *The Solution:* I developed a Flutter/Dart application with Riverpod state management and an `AppRepository` abstraction interfacing directly with an embedded SQLite engine. Synchronous in-memory caching guarantees zero UI stutter, while row mutations persist asynchronously. I integrated biometric authentication, background local scheduled notifications, and Android OS `FLAG_SECURE` window protections.
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4. Leadership Maturity, Operations & Personality

My technical competencies are reinforced by a well-rounded, mature personality shaped by real-world operational leadership. Between 2022 and 2023, alongside software development, I served as **Operations & Systems Lead** in hospitality management in Addis Ababa.

In this capacity, I directed day-to-day property logistics, guest relations, procurement, and front-desk administration, while modernizing legacy manual ledgers with computerized POS and digital booking workflows. This experience instilled in me:

- **High Emotional Intelligence & Crisis Leadership:** Navigating intense customer escalations, negotiating with commercial suppliers, and leading multi-disciplinary staff teams under tight deadlines.
- **Financial Accountability:** Managing real revenue streams, daily reconciliation, and inventory auditing where errors directly impact profitability.
- **Resilience & Resourcefulness:** Solving mission-critical bottlenecks in emerging-market infrastructure where power, internet, or logistics are disrupted.

This unique combination of software engineering discipline and operational leadership gives me the maturity, communication skills, and adaptability required to excel in collaborative European postgraduate research laboratories.

5. Why France & Academic Project (*Projet d'Études*)

France is an extraordinary destination for advanced computer science education. French higher education harmonizes mathematical formalism with rigorous industrial engineering, producing computer scientists who excel not only in writing software, but in proving its correctness, scalability, and safety.

I am targeting Master of Science programs in Computer Science (M1/M2) at leading French institutions renowned for systems engineering and distributed networks, including:

- **Université Paris-Saclay:** Master in Computer Science — *Distributed Systems and Cloud Computing / Foundations of Computer Science.*

- **Institut Polytechnique de Paris (IP Paris - Télécom Paris / École Polytechnique):** Master of Science in Computer Science — *Parallel and Distributed Systems*.
- **Université Grenoble Alpes (UGA):** Master in Computer Science (MoSIG) — *Software Architecture & Dependable Systems*.
- **Université de Lille / Inria Nord Europe:** Master Informatique — *Distributed Software Systems and Networks*.
- **Université Côte d'Azur / Inria Sophia Antipolis:** Master in Computer Science & Cloud Systems.

During my Master's studies in France, I intend to focus on:

1. **Formal Verification & Distributed Consensus:** Studying Raft, Paxos, and Byzantine Fault Tolerant (BFT) protocols to build software that maintains deterministic consistency across untrusted networks.
2. **Cloud-Native & Edge Orchestration:** Researching containerization, microservices communication patterns, and decentralized compute pipelines suited for low-bandwidth, high-latency edge environments.
3. **Dependable Software Engineering:** Utilizing advanced static analysis, formal methods, and automated testing frameworks pioneered by French institutions.

6. Long-Term Career Vision & Eiffel Fellowship Justification

My five-year professional roadmap consists of two sequential phases:

- **Phase 1 (Post-Graduation in France / Europe):**

Following graduation, I plan to leverage the French 2-year *Recherche d'emploi / Création d'entreprise* (APS) residence authorization to work as a **Distributed Systems Engineer / Cloud Architect** in France's thriving tech ecosystem (e.g., in Paris or Sophia Antipolis). Contributing directly to enterprise cloud infrastructure, high-throughput financial distributed ledgers, or open-source software will solidify my mastery of industrial-grade systems engineering.

- **Phase 2 (Long-Term Leadership & Tech Innovation in Africa):**

In the long term, I aim to return to Ethiopia and East Africa as a **Principal Software Architect and Technology Leader**. As African economies undergo rapid digitization—from the launch of capital markets and digital national identity systems (Fayda) to pan-African cross-border payment switches—there is an acute need for engineers who understand how to design resilient, secure, and fault-tolerant public digital infrastructure.

The **France Excellence Eiffel Scholarship** would be transformative for my academic trajectory. By eliminating financial constraints, it will allow me to immerse myself fully in coursework, laboratory research, and French academic life, while serving as a committed cultural ambassador between Ethiopia and the French Republic.

I approach this opportunity with deep intellectual curiosity, demonstrated technical competence, and an unwavering commitment to academic excellence. I thank the admissions committee and the Campus France selection panel for considering my application.